Anthony Tranduc

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Experience

Amazon Robotics — Boston, MA
Software Development Engineer II — *April 2024 – Present*
- Lead two engineers building a controls-in-the-loop simulator on Unreal Engine with USD-based assets, integrating production PLC and SoftPLC software to validate logic against virtual hardware
- Architected a testing framework that made the simulation programmatically observable and controllable for the first time – previously limited to a few MQTT spawn topics – giving a 5-engineer customer team their own on-demand virtual test environments where a single physical testbed had been the only option
- Designed the framework's API surface to be agent-friendly, cutting delivery of a new test API from concept to shipped to a single day, with autonomous agents driving implementation, testing, and adversarial review under human oversight
- Built distributed digital twins of fulfillment centers, running 4-floor warehouses across tens of containers with multiple experiments in parallel

Software Development Engineer I — *August 2022 – April 2024*
- Built a software-in-the-loop framework that runs teams' production software unmodified in simulation, replacing a legacy system that required intrusive changes to every integrating service
- Developed workcell models that mock real hardware, letting software teams integrate and test against virtual systems before physical deployment

Tufts University Department of Computer Science — Medford, MA
Algorithms Teaching Assistant — *January 2020 – May 2022*
- Tutored students on analyzing and applying algorithms during weekly office hours
- Graded 50+ homework assignments weekly

Amazon Robotics — North Reading, MA
Software Development Engineer Intern — *May 2021 – August 2021*
- Built a proof-of-concept that shaped the team's 2022 roadmap, piping simulation time into a live containerized service by intercepting its time calls through a preloaded library
- Built tooling to scale Amazon Robotics' simulation infrastructure using Docker, RabbitMQ, C, Python, and Java

Tufts JumboCode — Medford, MA
Technical Lead — *September 2020 – June 2021*
- Led a 12-developer team building a medical resource finder web application
- Reviewed 3–5 pull requests weekly and mentored developers on technical decisions

Codio — Cambridge, MA
Content and Code Development Intern — *June 2020 – August 2020*
- Developed auto-grading API for Processing programs using an abstract syntax tree
- Created ~100 data structures and algorithms assessments in C++, Java, and Python

Education

Tufts University — Medford, MA
Bachelor of Science in Computer Science, Philosophy — *Class of 2022*
GPA: 3.82/4.00, Summa Cum Laude, Phi Beta Kappa

Technical Skills

Languages: C++, Python, Java
AI & Agentic Workflows: AI-assisted development, agentic workflows, multi-agent systems, autonomous agent loops, LLM tooling (Claude Code, opencode)
Tooling & Messaging: Docker, Linux, Git, MQTT, RabbitMQ
Simulation & Robotics: Unreal Engine, USD, digital twins, discrete event simulation, distributed simulation